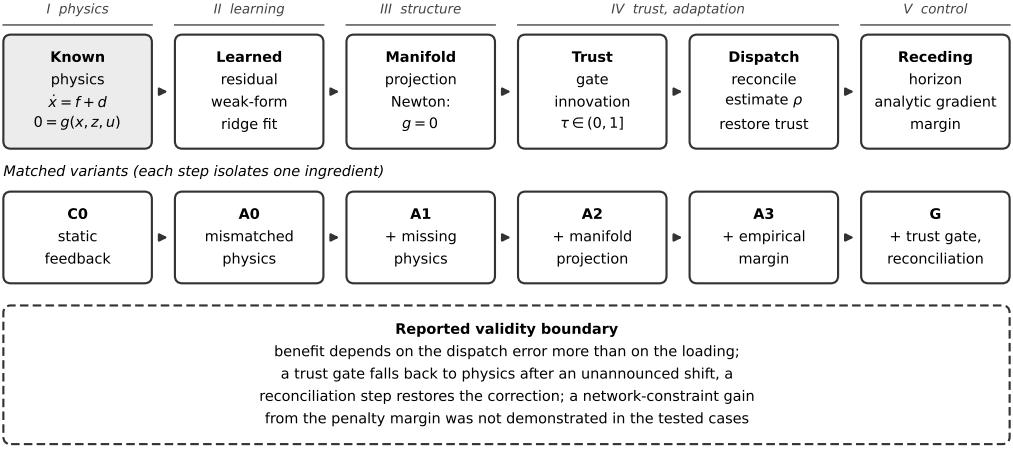
I physics
II learning
III structure
IV trust, adaptation
V control
Known
physics
$\dot{x} = f + d$
$0 = g(x, z, u)$
Learned
residual
weak-form
ridge fit
Manifold
projection
Newton:
$g = 0$
Trust
gate
innovation
$\tau \in (0, 1]$
Dispatch
reconcile
estimate ρ
restore trust
Receding
horizon
analytic gradient
margin
Matched variants (each step isolates one ingredient)
C0
static
feedback
A0
mismatched
physics
A1
+ missing
physics
A2
+ manifold
projection
A3
+ empirical
margin
G
+ trust gate,
reconciliation
Reported validity boundary
benefit depends on the dispatch error more than on the loading;
a trust gate falls back to physics after an unannounced shift, a
reconciliation step restores the correction; a network-constraint gain
from the penalty margin was not demonstrated in the tested cases